INSTITUT UNIVERSITAIRE DES SCIENCES

Faculté des sciences techniques (FST)

L3

TD3 Réseaux

Nom : Isidor

Prénom : Adner Rubens

Niveau : L3

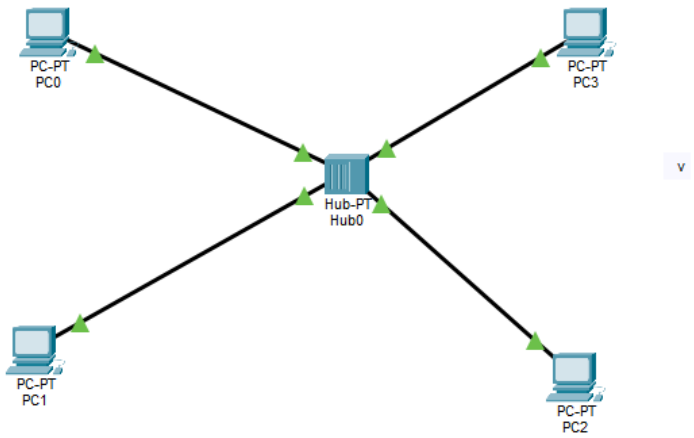
Date :08/11/2026

Objectif du TD

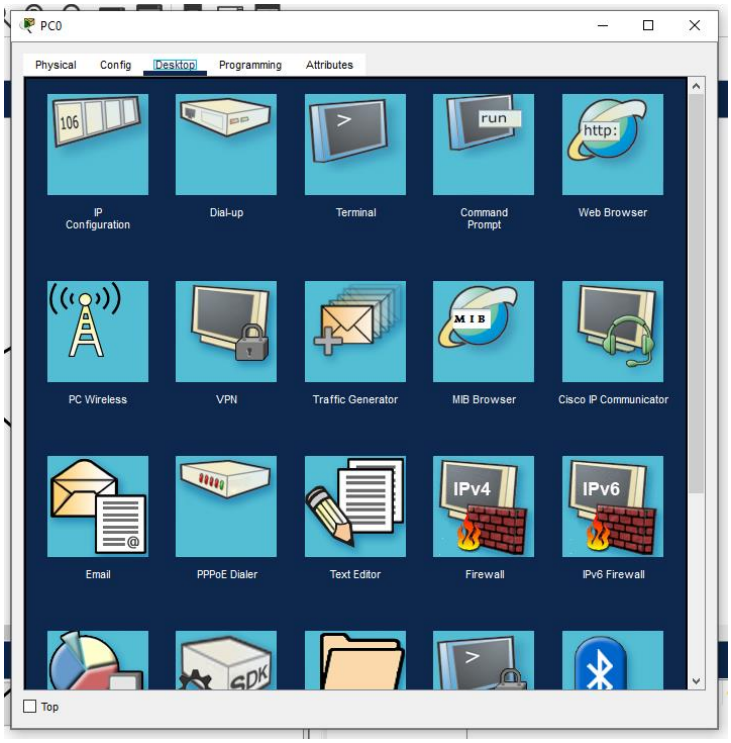
1- Savoir attribuer des adresses IP valides aux machines.
2- Comprendre l'adresse IPv4 et IPv6
3- Configurer des adresses IP sur des hôtes et routeurs dans Cisco Packet Tracer.
4- Vérifier la connectivité avec les commandes ping et ping ipv6.

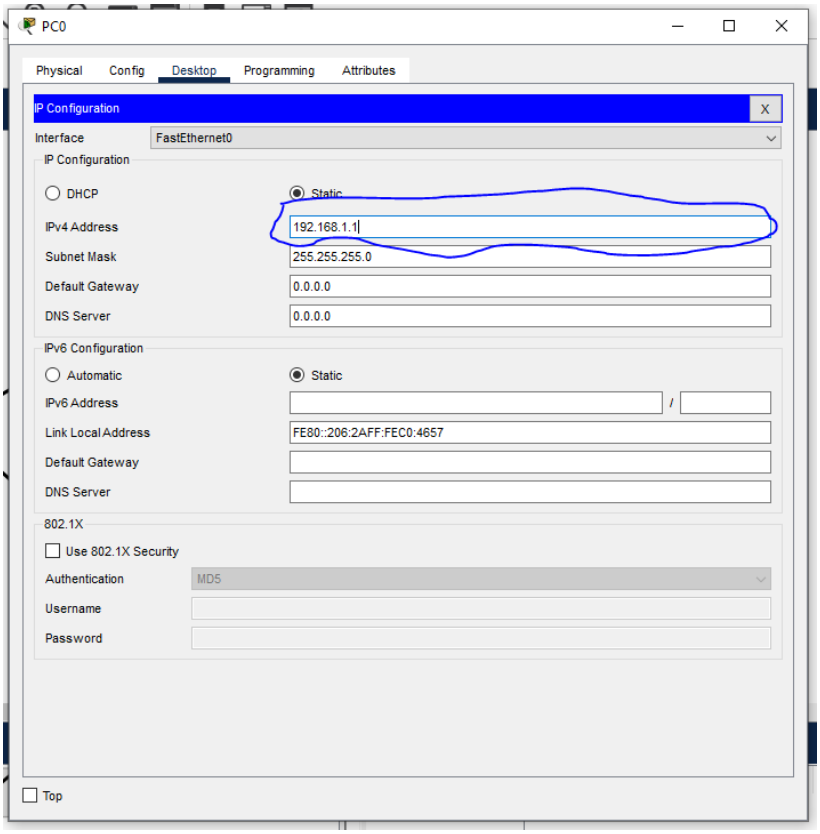
1- Créer une topologie avec :

- Un Hub
- 4 PC connectés au switch

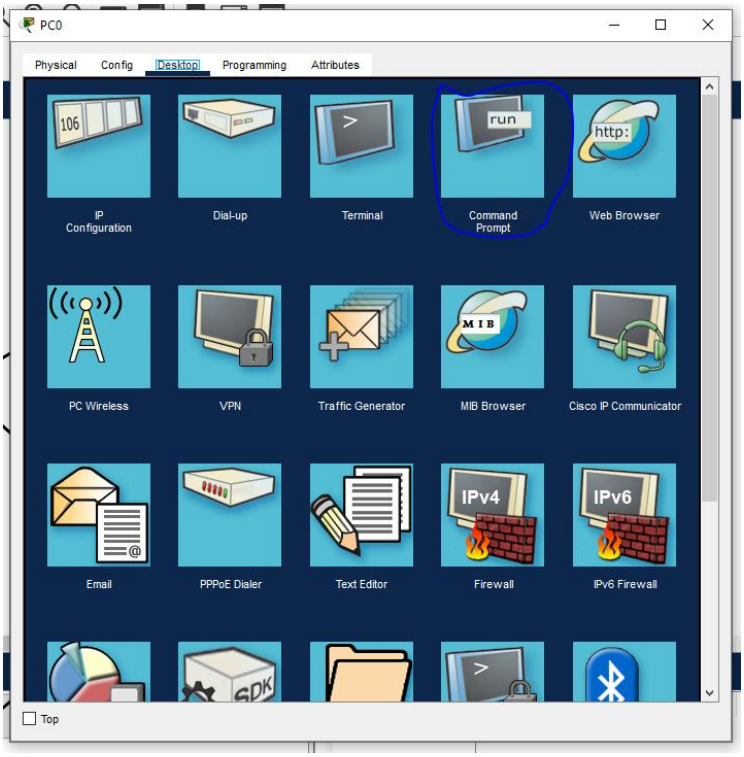


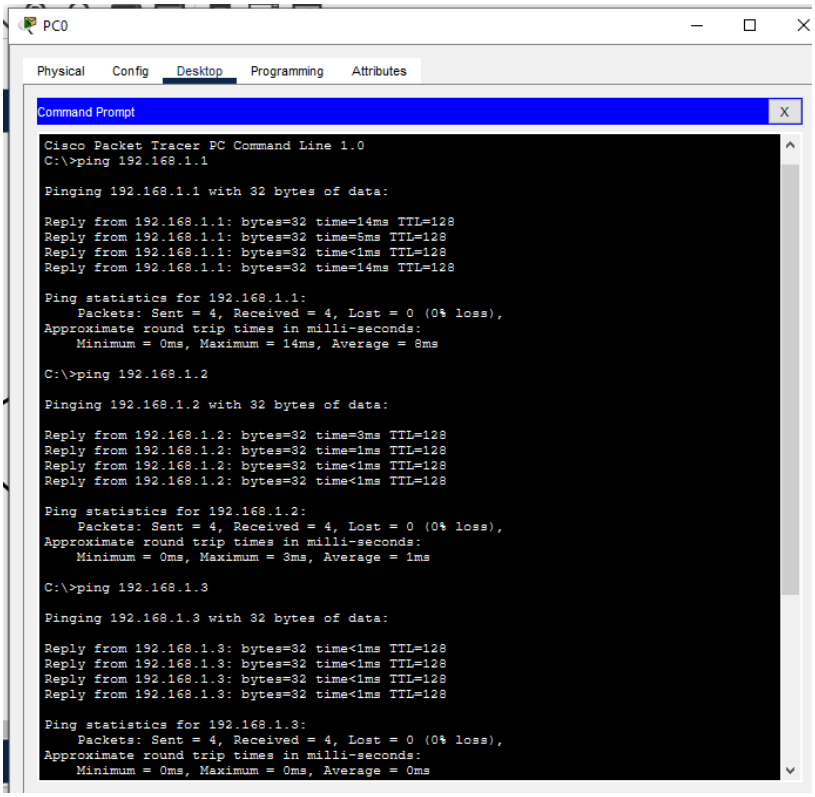
2- Configurez les adresses IPv4.

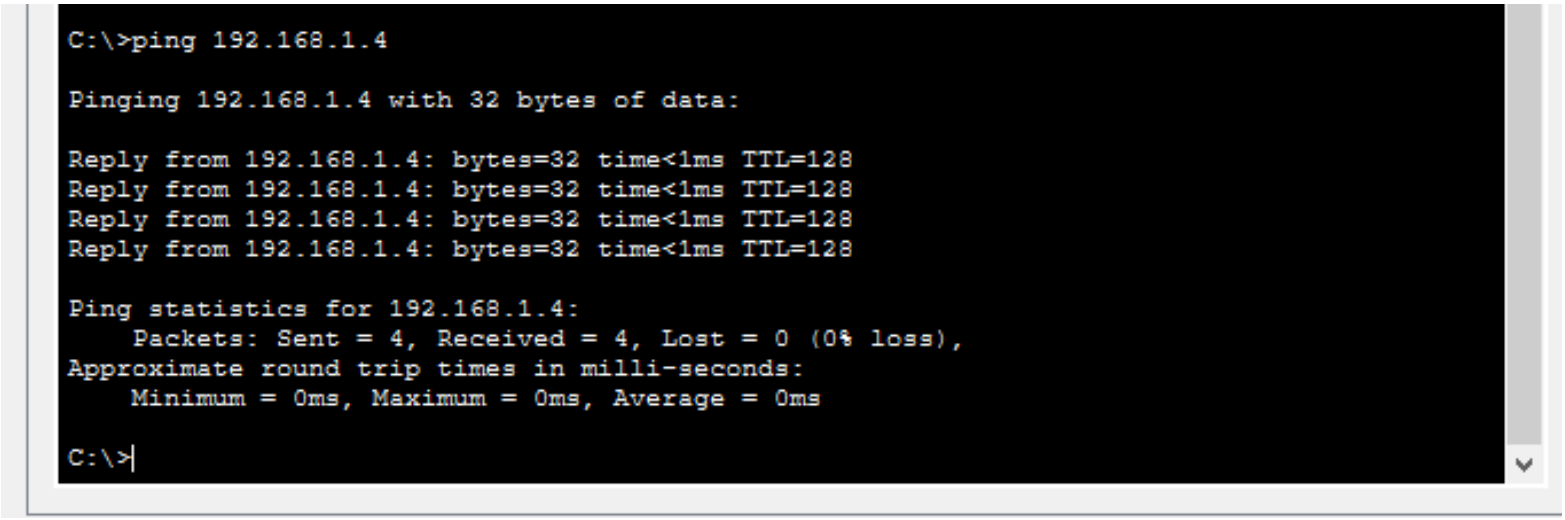




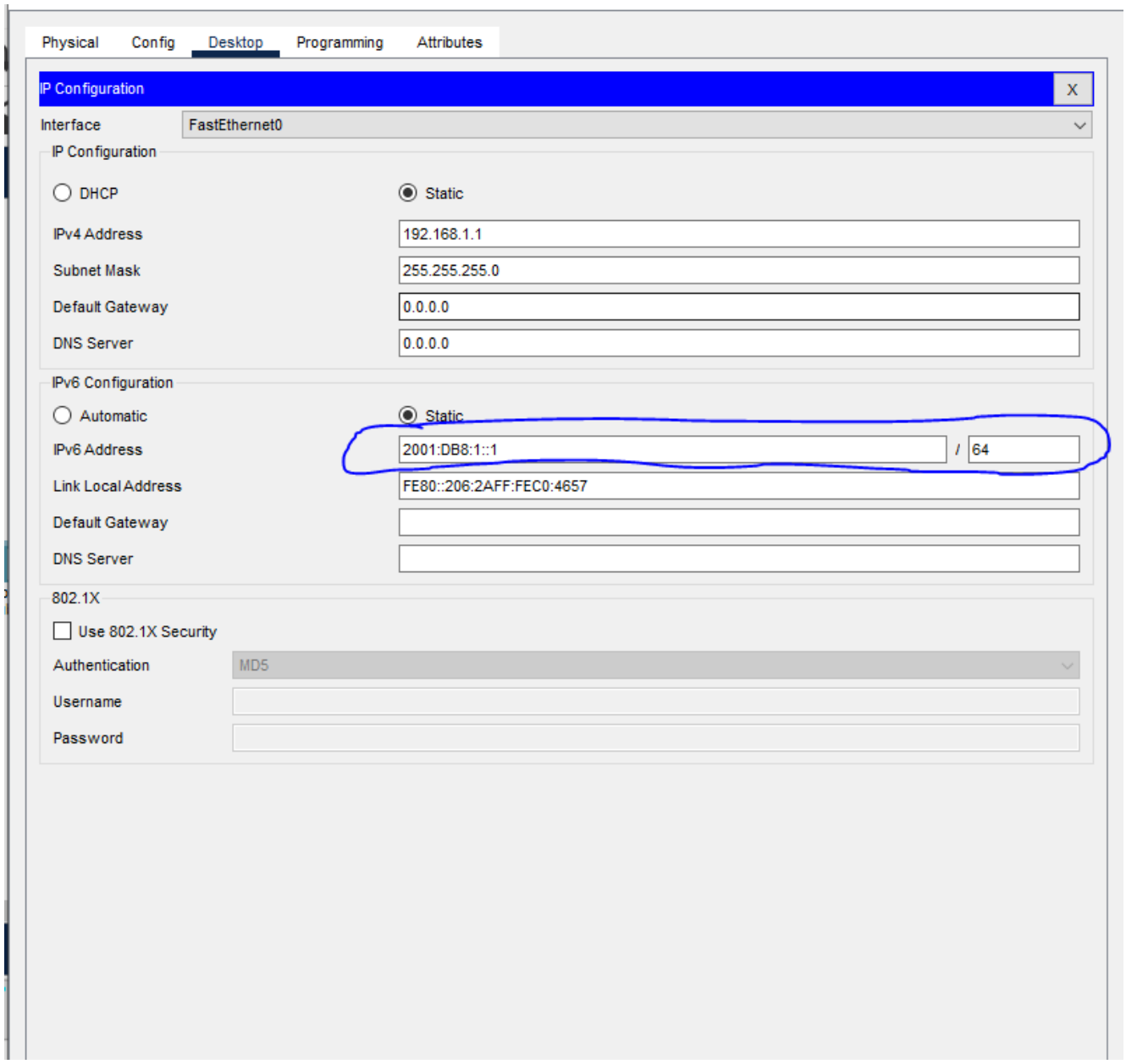
3- Vérifier la connectivité en faisant ping entre tous les PC.







4- Dans la même topologie configurez également une adresse IPv6 pour chaque PC.



5- Testez la connectivité

```
C:\>ping 2001:db8:1::1

Pinging 2001:db8:1::1 with 32 bytes of data:

Reply from 2001:DB8:1::1: bytes=32 time=6ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time=15ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time=14ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time=14ms TTL=128

Ping statistics for 2001:DB8:1::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 6ms, Maximum = 15ms, Average = 12ms

C:\>ping 2001:db8:1::2

Pinging 2001:db8:1::2 with 32 bytes of data:

Reply from 2001:DB8:1::2: bytes=32 time=4ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time<1ms TTL=128

Ping statistics for 2001:DB8:1::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 4ms, Average = 1ms
```

```
C:\>ping 2001:db8:1::3

Pinging 2001:db8:1::3 with 32 bytes of data:

Reply from 2001:DB8:1::3: bytes=32 time=4ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128

Ping statistics for 2001:DB8:1::3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 4ms, Average = 1ms

C:\>ping 2001:db8:1::4

Pinging 2001:db8:1::4 with 32 bytes of data:

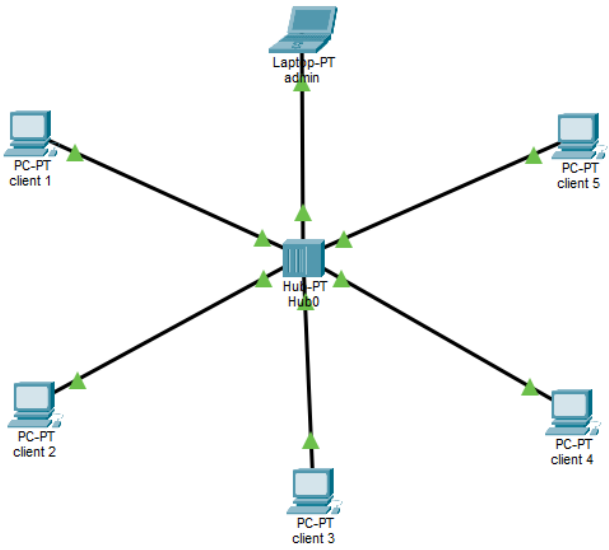
Reply from 2001:DB8:1::4: bytes=32 time=4ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128

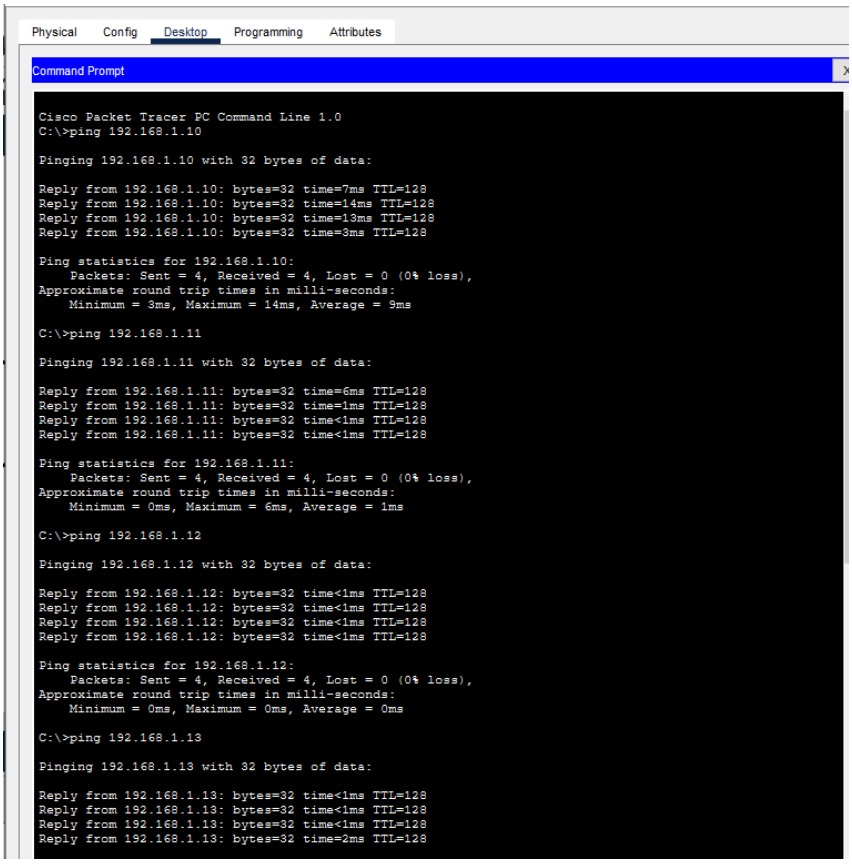
Ping statistics for 2001:DB8:1::4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 4ms, Average = 1ms

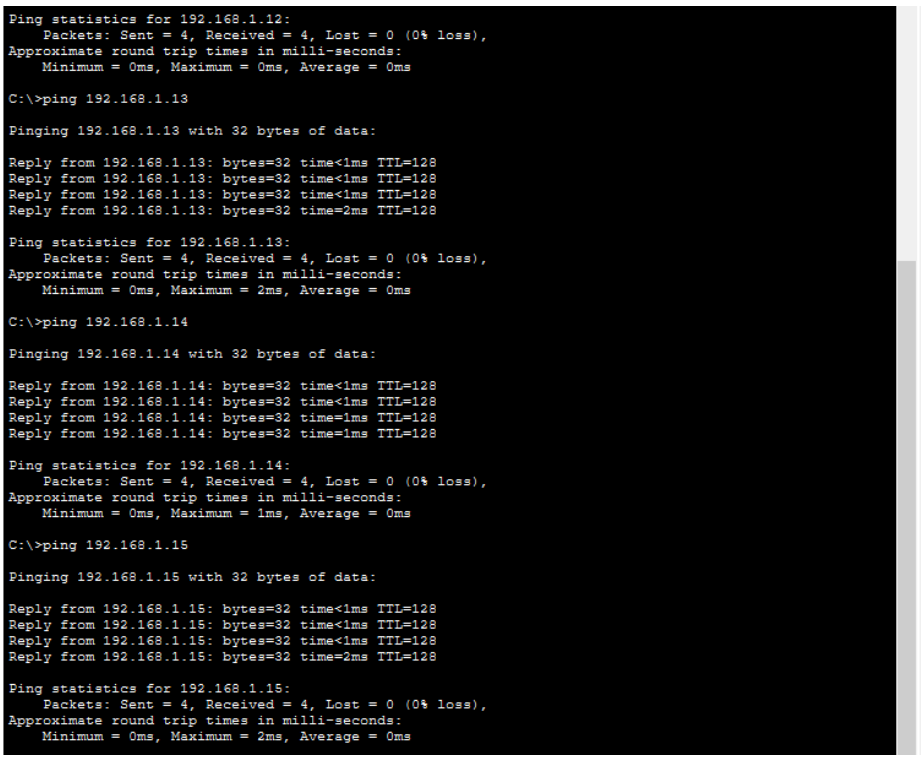
C:\>
```

PARTIE 2

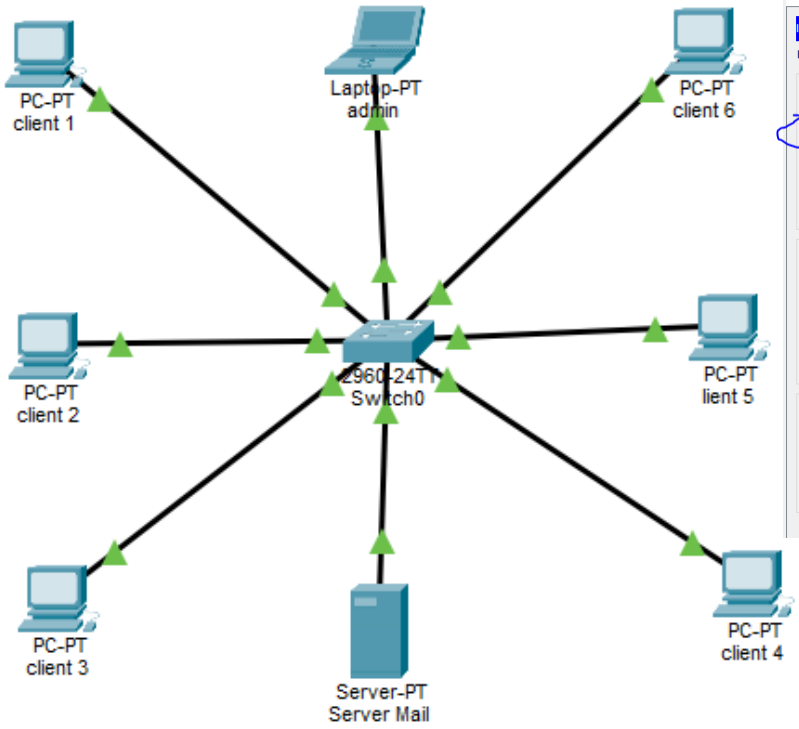
1- Reproduisez cette topologie en configurant les adresses IPv4 indiqué dans le tableau ci-dessus puis vérifiez la connectivité.

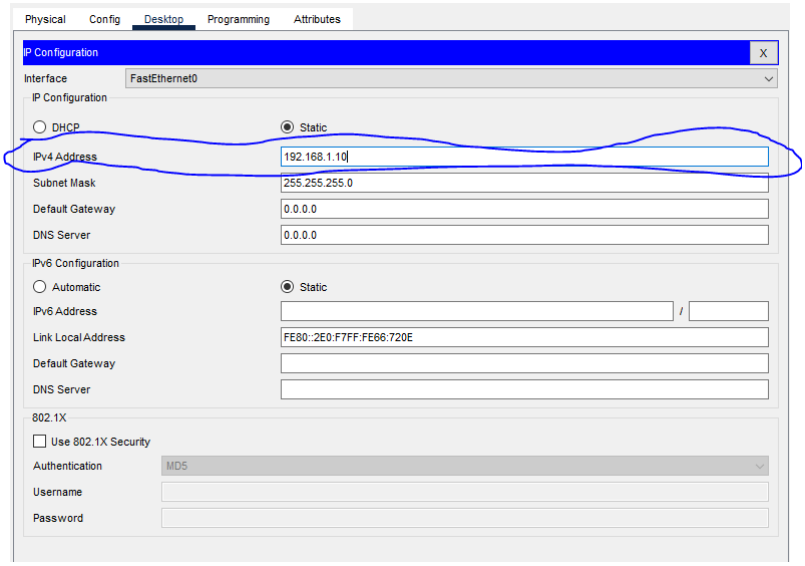


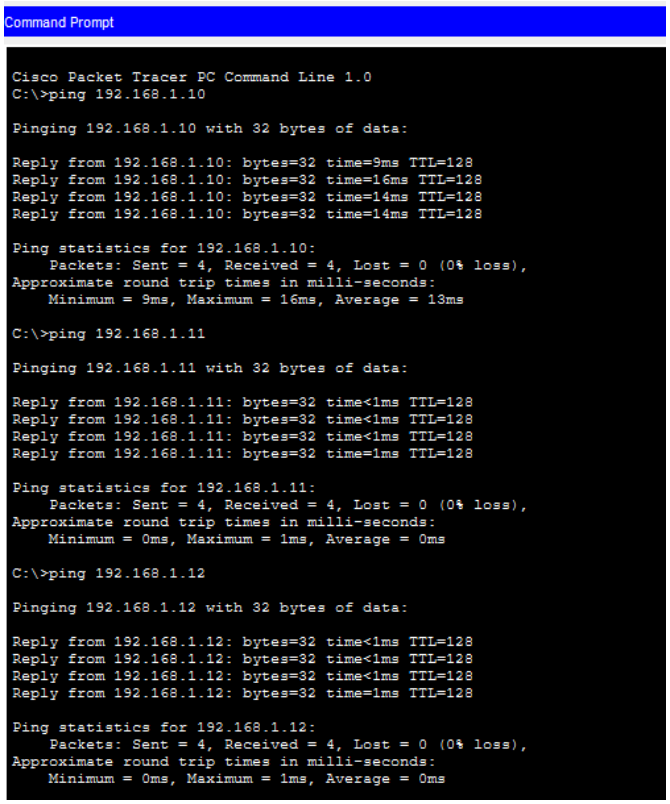




2- Reproduisez cette topologie en configurant les adresses IPv4 indiqué dans le tableau ci-dessus puis vérifiez la connectivité.







```
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time=9ms TTL=128
Reply from 192.168.1.10: bytes=32 time=16ms TTL=128
Reply from 192.168.1.10: bytes=32 time=14ms TTL=128
Reply from 192.168.1.10: bytes=32 time=14ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 9ms, Maximum = 16ms, Average = 13ms

C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.1.12

Pinging 192.168.1.12 with 32 bytes of data:

Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

```
C:\>ping 192.168.1.13

Pinging 192.168.1.13 with 32 bytes of data:

Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time=1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.13:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.1.14

Pinging 192.168.1.14 with 32 bytes of data:

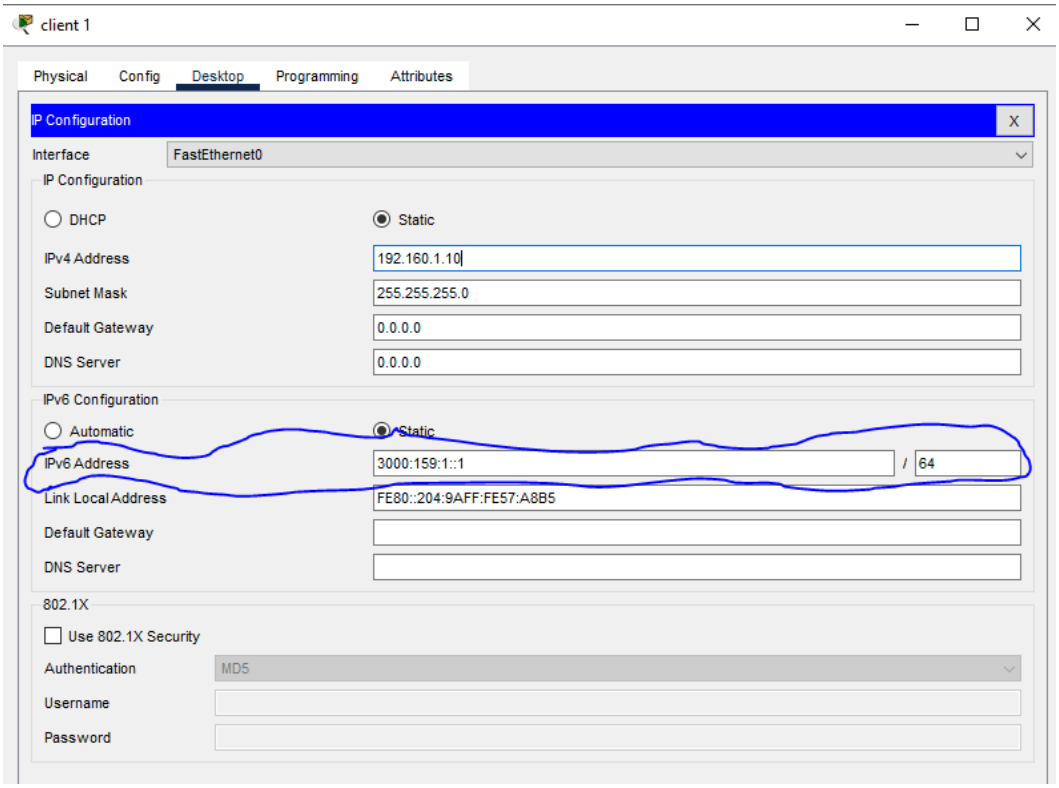
Reply from 192.168.1.14: bytes=32 time<1ms TTL=128
Reply from 192.168.1.14: bytes=32 time<1ms TTL=128
Reply from 192.168.1.14: bytes=32 time=10ms TTL=128
Reply from 192.168.1.14: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.14:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>ping192.168.1.15
Invalid Command.

C:\>
```

3- Configuration de l'adresse IPv6 pour chaque PC de la deuxième topologie.




```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 3000:159:1::1

Pinging 3000:159:1::1 with 32 bytes of data:

Reply from 3000:159:1::1: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::1: bytes=32 time=4ms TTL=128
Reply from 3000:159:1::1: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::1: bytes=32 time<1ms TTL=128

Ping statistics for 3000:159:1::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 4ms, Average = 1ms

C:\>ping 3000:159:1::2

Pinging 3000:159:1::2 with 32 bytes of data:

Reply from 3000:159:1::2: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::2: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::2: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::2: bytes=32 time<1ms TTL=128

Ping statistics for 3000:159:1::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>ping 3000:159:1::3

Pinging 3000:159:1::3 with 32 bytes of data:

Reply from 3000:159:1::3: bytes=32 time=10ms TTL=128
Reply from 3000:159:1::3: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::3: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::3: bytes=32 time<1ms TTL=128

Ping statistics for 3000:159:1::3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>ping 3000:159:1::4

Pinging 3000:159:1::4 with 32 bytes of data:

Reply from 3000:159:1::4: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::4: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::4: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::4: bytes=32 time<1ms TTL=128

Ping statistics for 3000:159:1::4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 3000:159:1::5

Pinging 3000:159:1::5 with 32 bytes of data:

Reply from 3000:159:1::5: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::5: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::5: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::5: bytes=32 time<1ms TTL=128

Ping statistics for 3000:159:1::5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```